

Week 16: Full System Integration Summary

Planned Deliverables (from thesis plan)

- Integrate all contracts
- Create end-to-end test scenarios
- Deploy the complete system to the local testnet
- Begin preliminary performance logging

What Was Built

1. Multi-Layer Test Suite

A comprehensive 3-layer test architecture was implemented:

Layer 1: Unit Tests (23 tests)

Location: pallets/content-rights/src/tests.rs + pallets/content-rights/src/mock.rs

- Mock runtime with pallet-nfts + pallet-content-rights
- Tests for all extrinsics: register, subscribe, renew, PPV, transfer, check_access
- Error path coverage (unauthorized, expired, not found)

Layer 2: XCM Simulator Integration Tests

Location: xcm-simulator-tests/

- Simulates relay chain + 2 parachains
- Tests cross-chain subscription, renewal, PPV, ownership transfer
- Verifies XCM message construction and execution
- No actual network required

Relevant commit: adcacc4

Layer 3: Zombienet E2E Tests

Location: tests/xcm-e2e/

- Live multi-chain tests against running Zombienet
- Tests the full pipeline: fund sovereign accounts → send XCM → verify state changes
- Covers: cross-chain subscribe, renew, PPV, ownership transfer
- Uses polkadot-js API for direct chain interaction

Relevant commits:

- 7b61e08 — Zombienet XCM E2E tests (Layer 3)
- a754117 — E2E tests for all cross-chain XCM operations

2. Full System Deployment

The complete system runs on a 4-chain Zombienet topology:

```
Rococo Relay Chain (alice, bob)
+-- Bridge Hub (para 1013) -- Snowbridge beacon client, message routing
+-- AssetHub (para 1000) -- Ether foreign asset, reserve transfers
+-- Content Rights (para 100) -- pallet-content-rights, pallet-nfts,
                                pallet-revive
    ^
    |
    | Ethereum (Geth + Lodestar)
    | ^
    | |
    | | Gateway contracts (GatewayProxy, BeefyClient, etc.)
```

HRMP channels:

- Bridge Hub (1013) ↔ AssetHub (1000)
- AssetHub (1000) ↔ Content Rights (100)

3. End-to-End Test Scenarios

Table 1: End-to-End Test Scenarios

Scenario	Chains Involved	Status
Register content + subscribe (local)	Content Rights	Passing
Cross-chain subscription renewal	Content Rights → Content Rights (via relay)	Passing
Cross-chain PPV access	Content Rights → Content Rights (via relay)	Passing
Cross-chain ownership transfer	Content Rights → Content Rights (via relay)	Passing
Ethereum → Bridge Hub beacon relay	Ethereum → Bridge Hub	Passing
Ethereum → AssetHub token bridge	Ethereum → BH → AH	Passing (2 ETH bridged, verified)

4. Performance Logging

Preliminary performance data collected:

- XCM fees: ~75–100B tokens per cross-chain operation (due to BlockRatioFee proof_size scaling)
- Beacon finalization time: ~39 minutes with 8 validators on mainnet preset
- Local testnet block time: 12s (relay), 12s (parachains)

Deviations from Plan

Table 2: Deviations from Plan

Planned	Actual	Rationale
Single testnet topology	4-chain topology (Relay + BH + AH + CR)	Snowbridge requires Bridge Hub and AssetHub for Ethereum interop
Simple integration tests	3-layer test architecture (unit + XCM simulator + Zombienet E2E)	Comprehensive coverage across different abstraction levels
“Integrate all contracts”	Pallet + ink! contract + Snowbridge pipeline	The system is a pallet + contract + bridge architecture, not multiple contracts